

Clinical Risk Assessment Report: Patient 35

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 33.3% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.50x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Tumor Pixel Correlation (CT) (GLCM_Correlation.CT), which elevated the patient's absolute risk by +3.6%. Biologically, this feature measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 1.67 [Top 3%] (+3.6%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.18 [Avg (58th %ile)] (+3.4%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: 0.07 [Top 10%] (+2.6%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.28 [Avg (31th %ile)] (+2.3%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.67 [Avg (67th %ile)] (+2.1%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

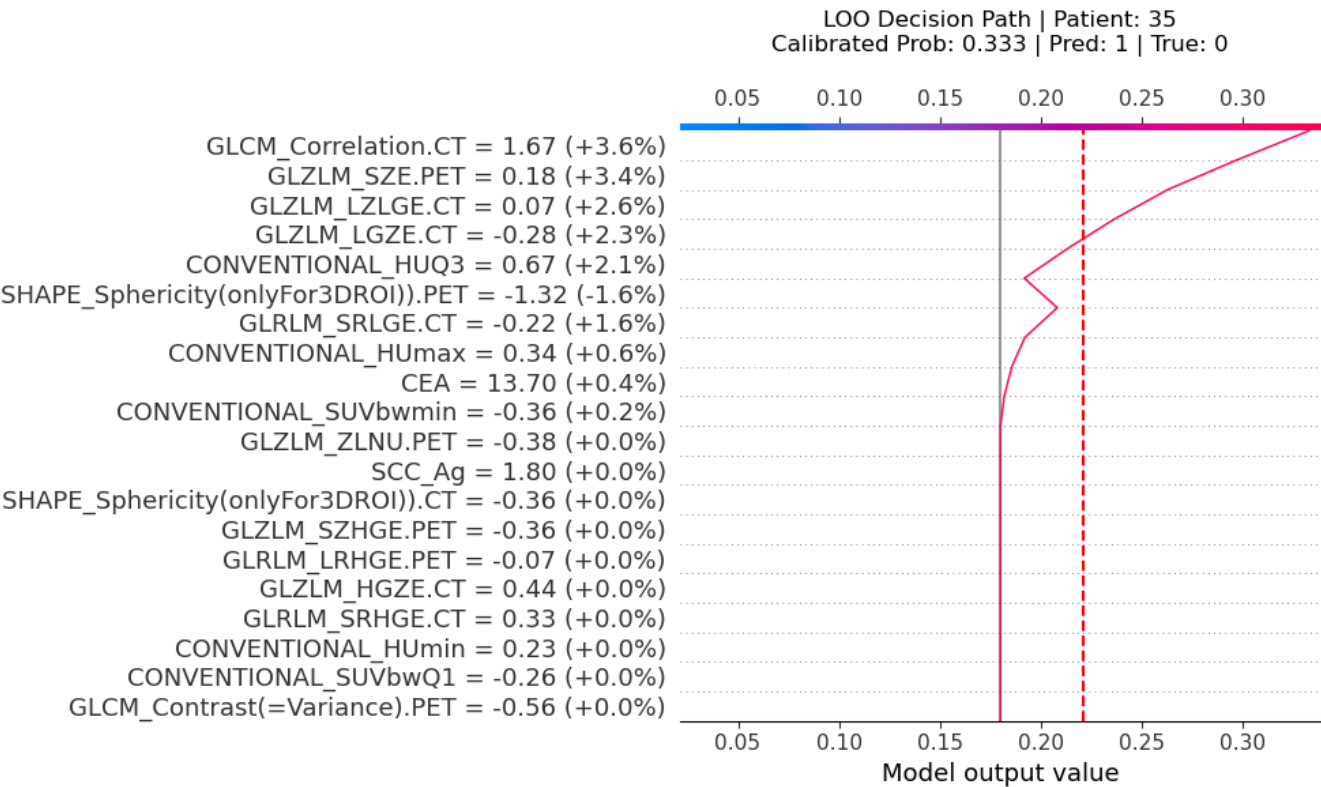
Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.22 [Bottom 12%] (+1.6%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.34 [Avg (66th %ile)] (+0.6%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

3. Protective / Mitigating Factors (Reducing Risk)

Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET **Value: -1.32 [Bottom 9%] (-1.6%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

